the diagnostic is computable before any knockoff is drawn

reference cohort
(covariates only, no response)

$\hat{\Sigma}$ = Ledoit-Wolf
shrinkage estimate

spectral diagnostic
λ_{min}, cond., PC1 share, effective rank

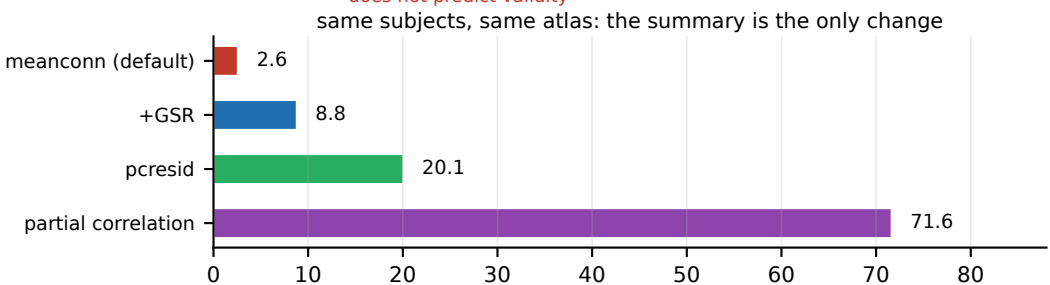
predicts power

main cohort
X, y

knockoff draw
$\tilde{X}$

W with penalty c
→ knockoff+ threshold
→ $\hat{S}$

condition 1
cohort mismatch $\leq \sim 0.12$
after subtracting
the sampling-noise floor

condition 2
penalty c satisfies
recovery ∧ convergence
at once (both unreported)

does not predict validity